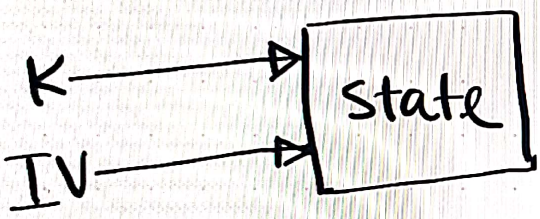
ChaCha 20 : Stream Cipher

Goal: generate deterministic, pseudorandom bits.

The ChaCha20 Keystream generator maintains a state.



The state is initialized with a secret key K and initialization vector IV.

during each iteration,
the Keystream generator generates a block of 64 bytes of Keystream. (512 bits)

To which a block of 64 bytes (512 bits) of plain text are added.

and then the state is updated.

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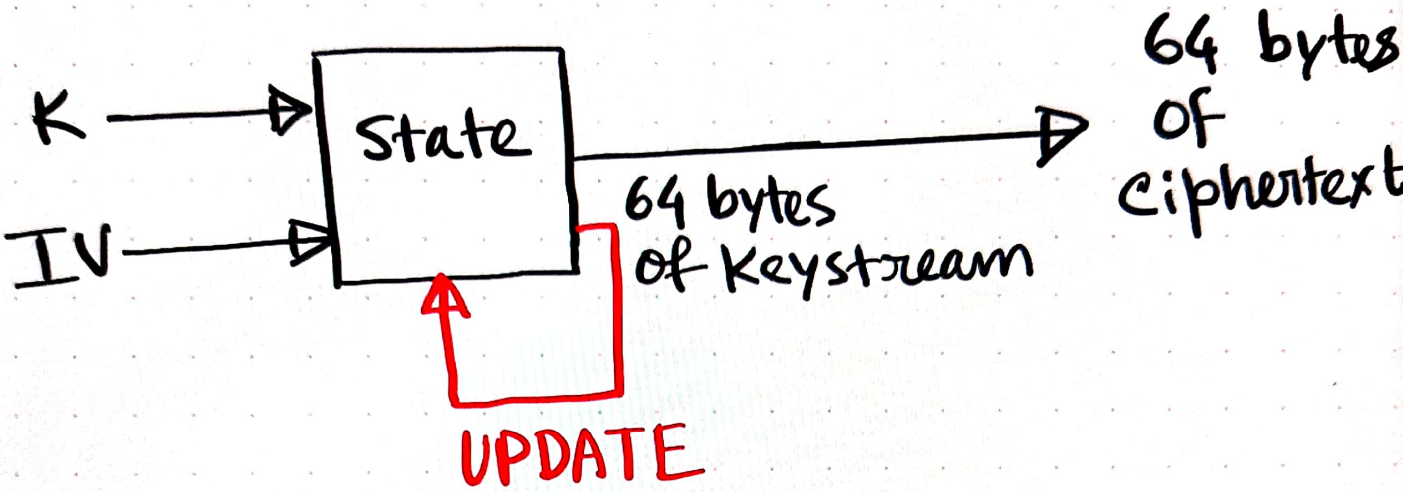
ChaCha20

Keystream Generator



The process is repeated until enough Key Stream bytes have been generated to encrypt the entire plaintext.

ChaCha 20 has 2 components,

(i) An initial State.

(ii) A Quater Round function.

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Description of the Initial State

Notation :

→ The Secret Key in ChaCha 20, is a randomly selected

256 bit Key $K = (K_1, K_2, K_3, \cdots, K_8)$

Where, K_i is a 32-bit word.
f_i is a 32-bit word.

→ 128 bit constant $f = (f_1, f_2, f_3, f_4)$

- A hexadecimal digit is a 4-bit number:

Binary	0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
Decimal	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Hexadecimal	0	1	2	3	4	5	6	7	8	9	*a*	*b*	*c*	*d*	*e*	*f*

- $f_1 = 0x61707865$, $f_2 = 0x3320646e$, $f_3 = 0x79622d32$, $f_4 = 0x6b206574$.

→ 96-bit nonce $n = (n_1, n_2, n_3)$

nonce is the initializing value IV
a non-repeating quantity.

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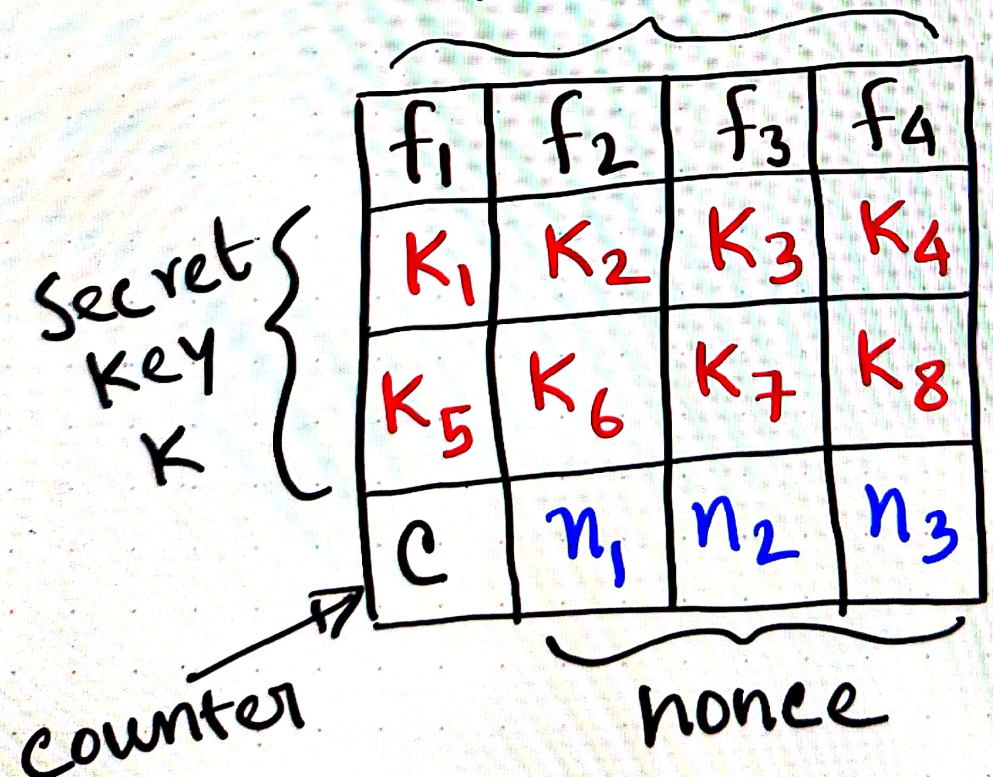
nonces are usually selected uniformly at random or by incrementing a counter.

→ 32 bit counter C.

The initial state of ChaCha20 can be represented by this MATRIX (4×4) of 32 bit words,



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f_1	f_2	f_3	f_4
K_1	K_2	K_3	K_4
K_5	K_6	K_7	K_8
C	n_1	n_2	n_3

=

S_1	S_2	S_3	S_4
S_5	S_6	S_7	S_8
S_9	S_{10}	S_{11}	S_{12}
S_{13}	S_{14}	S_{15}	S_{16}

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The QuaterRound function of ChaCha 20

it's called a quater round function, Since it operates on the 'quater' of the state.

→ On a binary string of len 128 bits

The input to the QRF consists of 4 32-bit words a, b, c, d.

→ The purpose of the QRF is to, mix up the bits of a, b, c, d in a complicated non-linear fashion.

This is done only by,

[Box-Sum] : $\boxplus$ int addition modulo 2^{32}

[XOR] : $\oplus$

[Left Rotation] : $\lll t$: left rotation by "t" bit positions.

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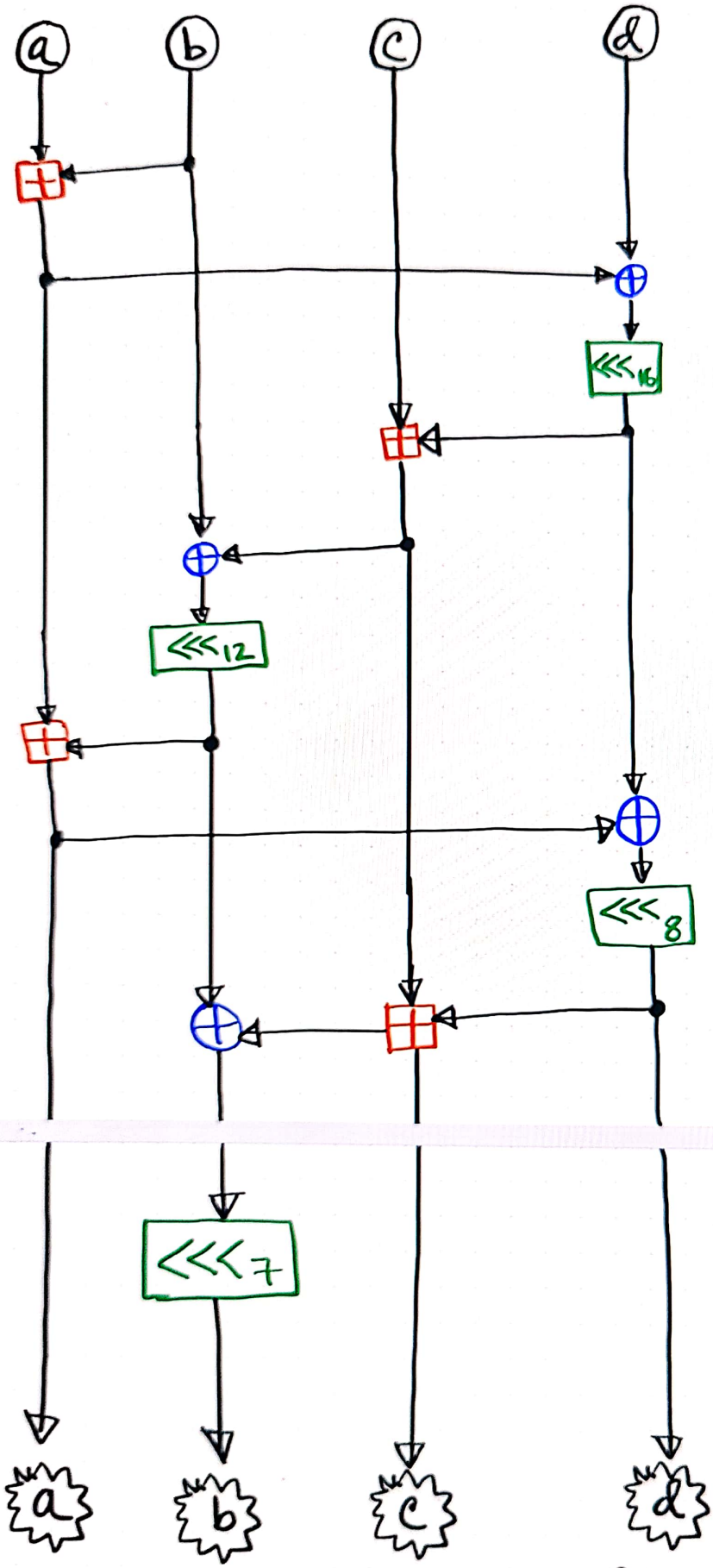
The result yeilded by above operations are all 32-bit words.



all horribly messed up. (diffusion)

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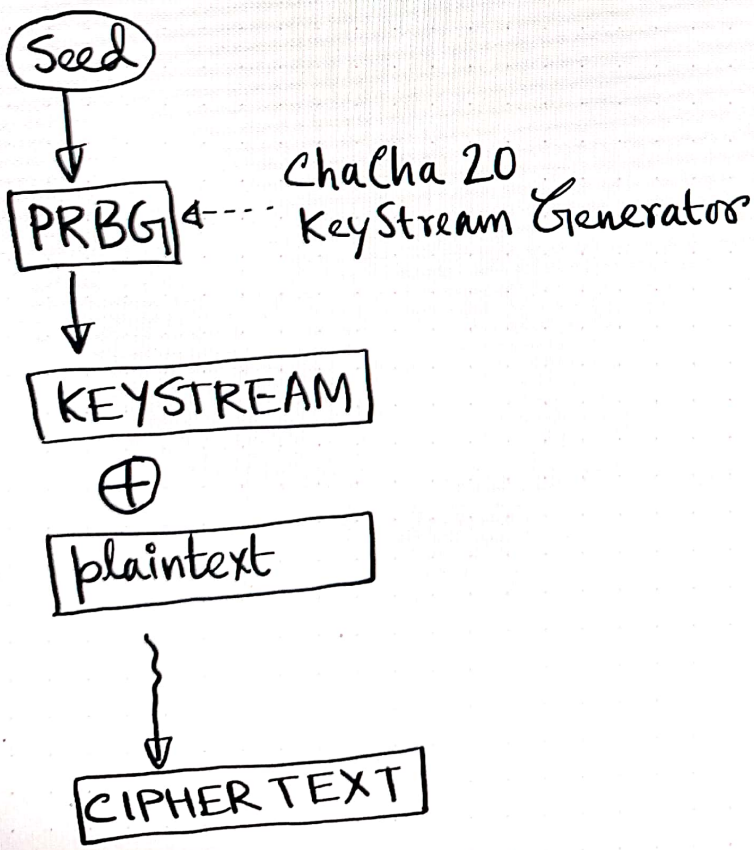
```
function QR(a,b,c,d):{
    a ← a ⊞ b ;          a ← a ⊞ b;
    d ← a ⊕ d ;          d ← a ⊕ b;
    d ← d <<<16;         d ← d <<<8

    c ← c ⊞ d ;          c ← c ⊞ d;
    b ← b ⊕ c;           b ← b ⊕ c;
    b ← b <<<12;         b ← b <<<7
    return a,b,c,d;
} end function ;
```


Description of the ChaCha 20

KeyStream Generator



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The KEYSTREAM GENERATOR of Chacha 20

Suppose, Alice wants to send Bob a message which is 64,000 byte in length.

Each iteration of the key stream generator outputs 64 Keystream bytes.

So, Alice would need to iterate this algorithm a 1000 times to generate all the 64,000 Keystream bytes.

// The counter is used the number of times the main-loop in the KeyStream Generator is used.

So, to encrypt the message m for Bob, Alice does the following,

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// with random nonce we ensure WHP that Alice never invokes the ChaCha20 KSG with the same inputs.

Select a nonce preferebly at random;

Initialize Counter ← 0 ;

While Keystream bytes are required do:

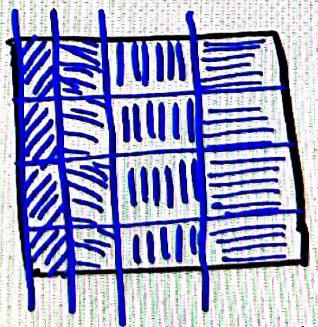
Create the initial state : S ←

f_1	f_2	f_3	f_4
K_1	K_2	K_3	K_4
K_5	K_6	K_7	K_8
C	n_0	n_1	n_2

S' ← S ; // S' is a copy of S.

UPDATE S by doing the following 10 times {

QR(S_1, S_5, S_9, S_{13});
QR(S_2, S_6, S_{10}, S_{14});
QR(S_3, S_7, S_{11}, S_{15});
QR(S_4, S_8, S_{12}, S_{16});



works on the coloumns.

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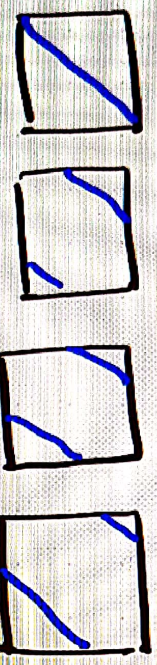
$QR(S_4, S_8, S_{12}, S_{16})$; coloumns.

$QR(S_1, S_6, S_{11}, S_{16})$;

$QR(S_2, S_7, S_{12}, S_{13})$;

$QR(S_3, S_8, S_9, S_{14})$;

$QR(S_4, S_5, S_{10}, S_{15})$;



} Works on diagonals of various type

// at this point the bits are combined in a quite complicated and non-linear way. This we'are doing 10 times!

} return $S \oplus S'$ (64 Keystream bytes 512 bit)

Counter: $C \leftarrow C+1$;

ENCRYPTION: The Keystream bytes are XORed with the plaintext bytes to produce the ciphertext bytes.

The nonce is appended to the Ciphertext.

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the nonce is appended

DECRYPTION: Upon receiving the nonce appended in the ciphertext, Bob regenerates the Keystream with the same Key and nonce and XORs the Keystream bits with the ciphertext bits.

- Why ChaCha20 is very efficient?
- To generate 64 Keystream bytes, The QR function is invoked 80 times. each invokation requires, 12 arithmetic operations.

So, 64 Keystream bytes can be generated only $12 \times 80 = 960$ many computer instructions.

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These operations, ARX are also built into the machine at the circuit level So they are all O(1) operations.

Chacha20 incorporates a { nonce and counter } for Keystream generation.

By this it ensures a Keystream is never reused even though the same Sec. Key be used for some period of time.

Since we use a 96 bit nonce for each plaintext that they encrypt and they increment the counter for each 64 byte Keystream block used to encrypt, for A 64 byte block of a particular plaintext, with very high probability, Chacha20 Keystream is never reused by Alice and Bob.

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for Key Stream generators, we need

(i) INDISTINGUISHABILITY.

(ii) UNPREDICTABILITY.

ChaCha family of Stream Ciphers

One approach of building PRGs is using Blum-Micali paradigm. The other builds PRGs from a more versatile primitive called 'Pseudorandom functions' in Counter mode.

PRGs underlying chacha stream cipher.

They take as input

(i) a 256 bit seed

(ii) a 64 bit nonce.

for now, ignore nonce and set it to 0.

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reference : A Graduate Course in CRYPTOGRAPHY boneh & Soup.

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Remarks

→ The design of Chacha is highly parallelizable and can take advantage of multiple processor cores to speed up encryption.

→ it enables random access to the output blocks. The output block j can be computed w/o computing all previous blocks!

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